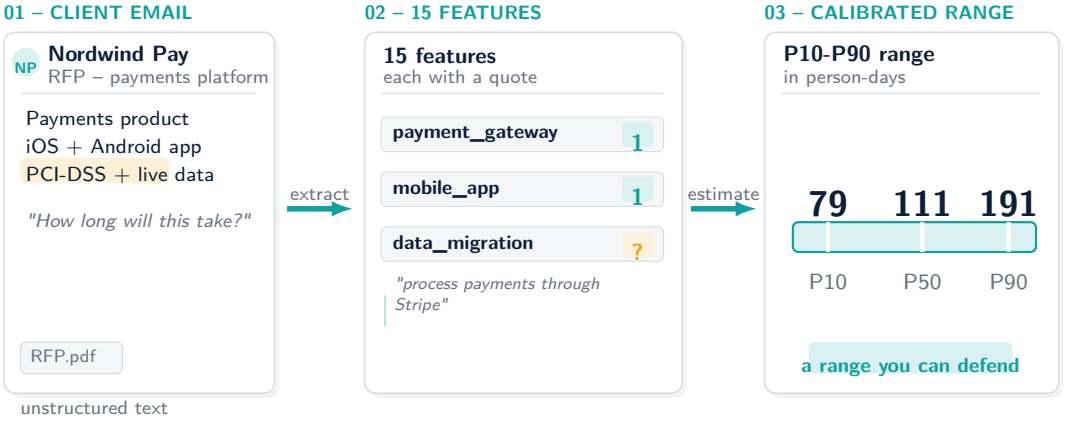
THE ESTIMATOR

From a client email to a range you can defend.

Nguyen My Thong · Anduin Cybernetics



A client email. One number. No evidence.

NP **Nordwind Pay** 09:14
Subject: Payments platform RFP

Hi — we need a payments product
for a mobile app on iOS and Android.
It requires PCI-DSS certification
and live updates on a dashboard.
No multi-currency for now.

— Elena Voss, Nordwind Pay

RFP_2026.pdf

How many days?

?

2 hours reading | 3 clarifying questions | **still a guess**

The LLM gives an answer. Not a defensible one.

LLM-only

258

days MAE
last 42 projects

The Estimator

19

days MAE
same 42 projects

no history | no calibration | no evidence | **missing info = guess**

What should the LLM decide?

Audit vs speed: only one option keeps a number we can trace.

A LLM end-to-end ✗

reads + estimates

- fast
- no audit
- no calibration

rejected

B LLM + rules ~

extract + rules

- auditable
- brittle
- no learning

trade-off

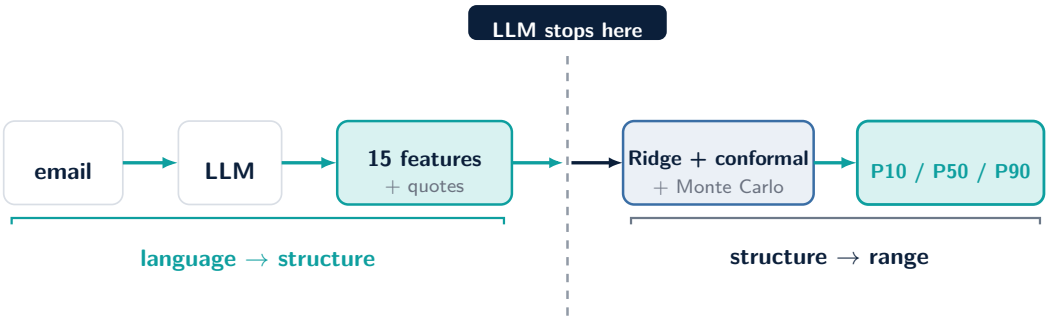
C LLM + math ✓

extract only → math

- auditable
- calibrated
- learns

chosen

So we split the job in two.



15 features. 3 states. A quote for each.

FEATURE	VALUE	EVIDENCE FROM THE EMAIL
`payment_gateway`	1	*"process payments through Stripe"*
`pci_dss`	1	*"requires PCI-DSS certification"*
`mobile_app`	1	*"mobile app for iOS and Android"*
`realtime_ui`	1	*"live updates on a dashboard"*
`multi_currency`	0	*"no multi-currency for now"*
`data_migration`	?	*not mentioned*

1 present 0 absent ? unknown

Example extraction

A range, not a point.



Example: Scenario A, offline run

Which model do we trust?

Coverage vs MAE: the simplest model still wins on trust.

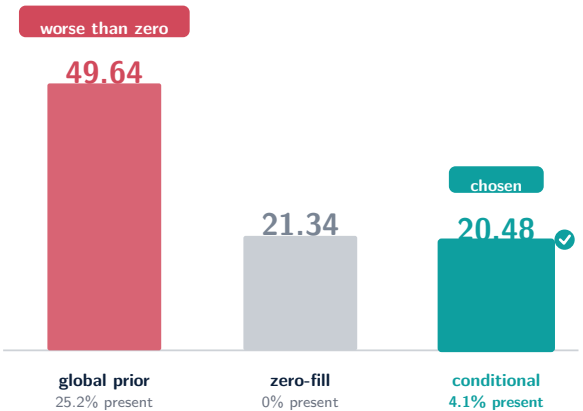
MODEL	MAE	COVERAGE	EXPLAINABLE	VERDICT
✓ Ridge + conformal	17.36	86.7%	yes	chosen — stable, inspectable
Split conformal	17.25	75.7%	yes	similar MAE, lower coverage
Gradient boosting	25.23	72.9%	no	overfits at 228 rows
Quantile GBM	25.56	66.5%	no	worse tails
Median baseline	32.76	77.1%	n/a	no learning

228 projects · walk-forward

lower MAE is better

What fills a missing feature?

Prior vs evidence: the mechanism decides the error, not the model.

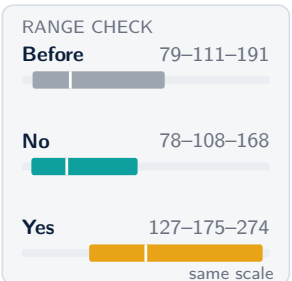




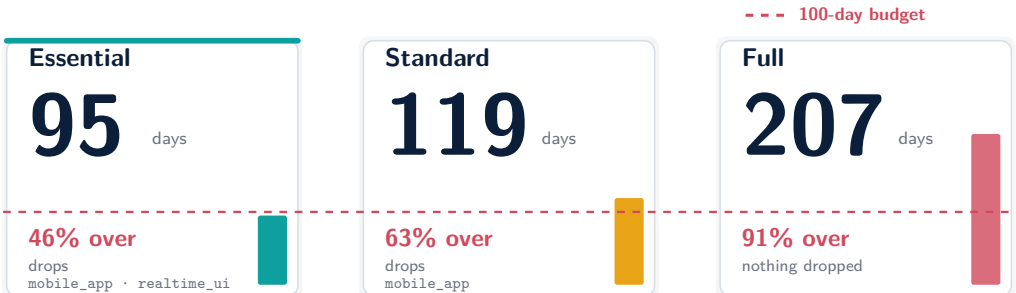
Ask what actually moves the number.

Migrate data from legacy systems?





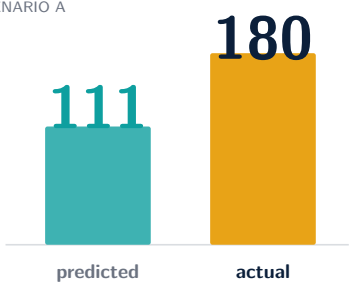
A budget turns the range into a decision.



Recommended **Essential** · renegotiate budget or timeline

Every finished project moves the model.

SCENARIO A

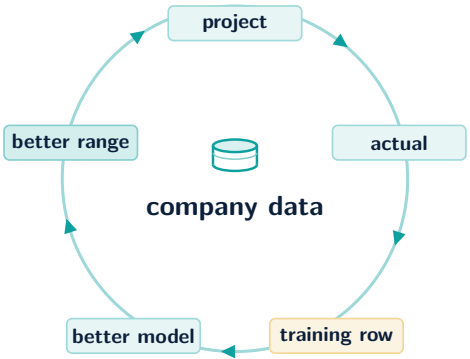


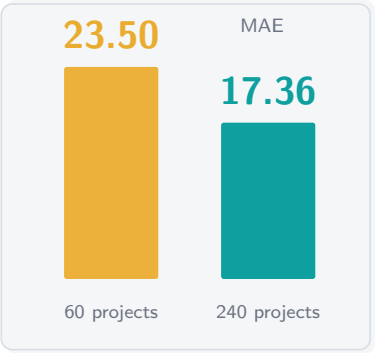
WEIGHTS	BEFORE	AFTER	DELTA
payment_gateway	1.49 →	1.51	+0.02
mobile_app	1.74 →	1.76	+0.02
pci_dss	1.50 →	1.52	+0.02

after drift · last 10 projects



More history, lower error.

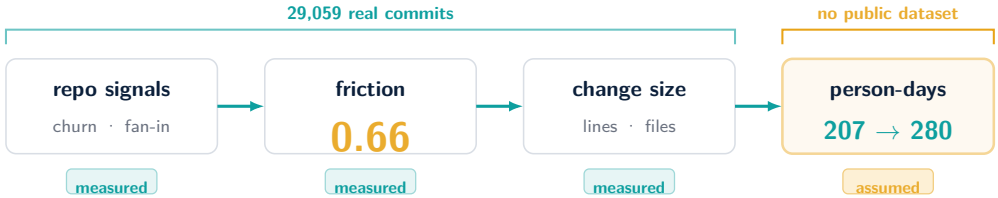




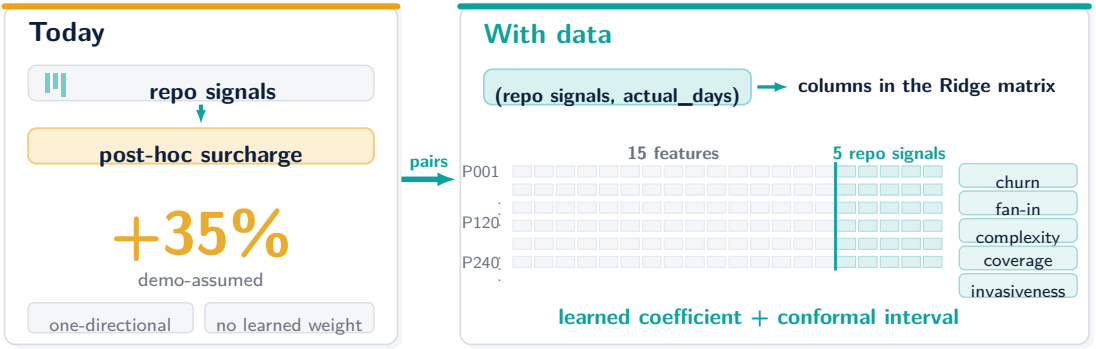
same model, more history

Measured, assumed, labelled.

Data vs assumption: three links have evidence, one does not.



When the data exists, the codebase becomes a feature.

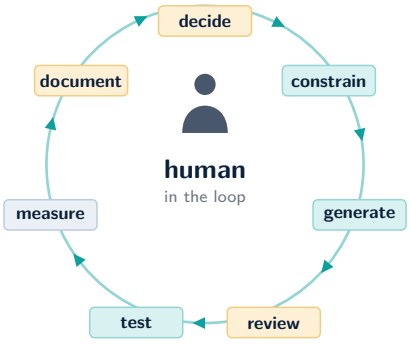


Vibe coding, with a seatbelt.

AI is fast. The human is the gatekeeper.

AI

- DeepSeek
- code assistant
- eval scripts



Human

- decide
- review
- gatekeep

built
on a phone

The tool stack, kept small.

Ceremony vs speed: four tools, and three deliberately refused.

Used

- **DeepSeek**
  extraction
- **Code assistant**
  boilerplate + tests
- **Eval scripts**
  measurement
- **Git history**
  29,059 commits

Deliberately not

- **50 MCP servers**
  setup for setup
- **10k-line skill files**
  unreadable
- **Autonomous agent**
  shipping code



No laptop. Same output.

Phone vs laptop: the weaker tool forced smaller, safer diffs.



what it forced

- small diffs
- explicit contracts
- tests as the safety net

The AI wrote plausible code. The review made it right.



Six checks before merge.

Fast vs trustworthy: generation is cheap, review is the work.

Review before merge

6 checks

- **Contract** — does it match the interface?
- **Failure modes** — empty, None, huge input?
- **Determinism** — same input → same output?
- **Evidence** — can I trace the number?
- **Tests** — fails when I break it?
- **Rollback** — can I revert in one commit?

What we chose not to build.

Scope vs critical path: four features deferred, not forgotten.

Guardrails

- 242 offline tests
- 31 decision entries
- no artifact → no claim
- graceful failure

Deliberately not built

auth / rate limit	next
extraction cache	not on the critical path
multi-language codebase	next
retrieval wiring	not on the critical path

The numbers.

17.36
MAE vs 32.93



86.7%
coverage 189/218



0.933
extraction F1



$0.20
for 240 emails

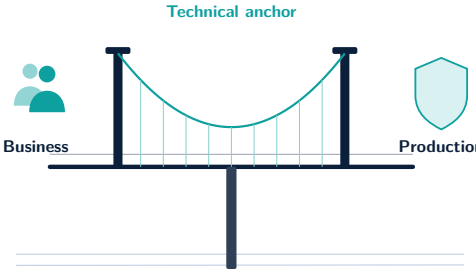


242 offline tests

31 decisions

20.48 MAE end-to-end

Business builds. Someone has to make it trustworthy.



the bridge, in practice

messy email	→ structured features
deterministic math	→ auditable number
questions	→ human review
actuals	→ learning loop

map → prototype → harden

I don't use AI to skip thinking.
I use it to think faster — then I test harder.

Thank you.